Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
# Update my R packages because was having errors with installing cowpot package.
#Then load libraries that was already install

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at /home/guest/EDE_FALL 2025
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
library(lubridate)


getwd()
```

```
## [1] "/home/guest/EDE_FALL 2025"
```

```r
NTL_LTER_PeterPaul <- read.csv(
  file = here("Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
  stringsAsFactors = TRUE)

Niwot_Ridge <- read.csv(
  file = here("Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),
  stringsAsFactors = TRUE)

# 2
# Reading dates as date format
NTL_LTER_PeterPaul$sampledate <- ymd(NTL_LTER_PeterPaul$sampledate)
Niwot_Ridge$collectDate <- ymd(Niwot_Ridge$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```r
#3
# Set the theme for the plots and graphs
my_theme <- theme_bw(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
    axis.title = element_text(size = 11, face = "italic"),
    panel.grid.minor = element_blank(),
    legend.position = "bottom"
  )

theme_set(my_theme)
```

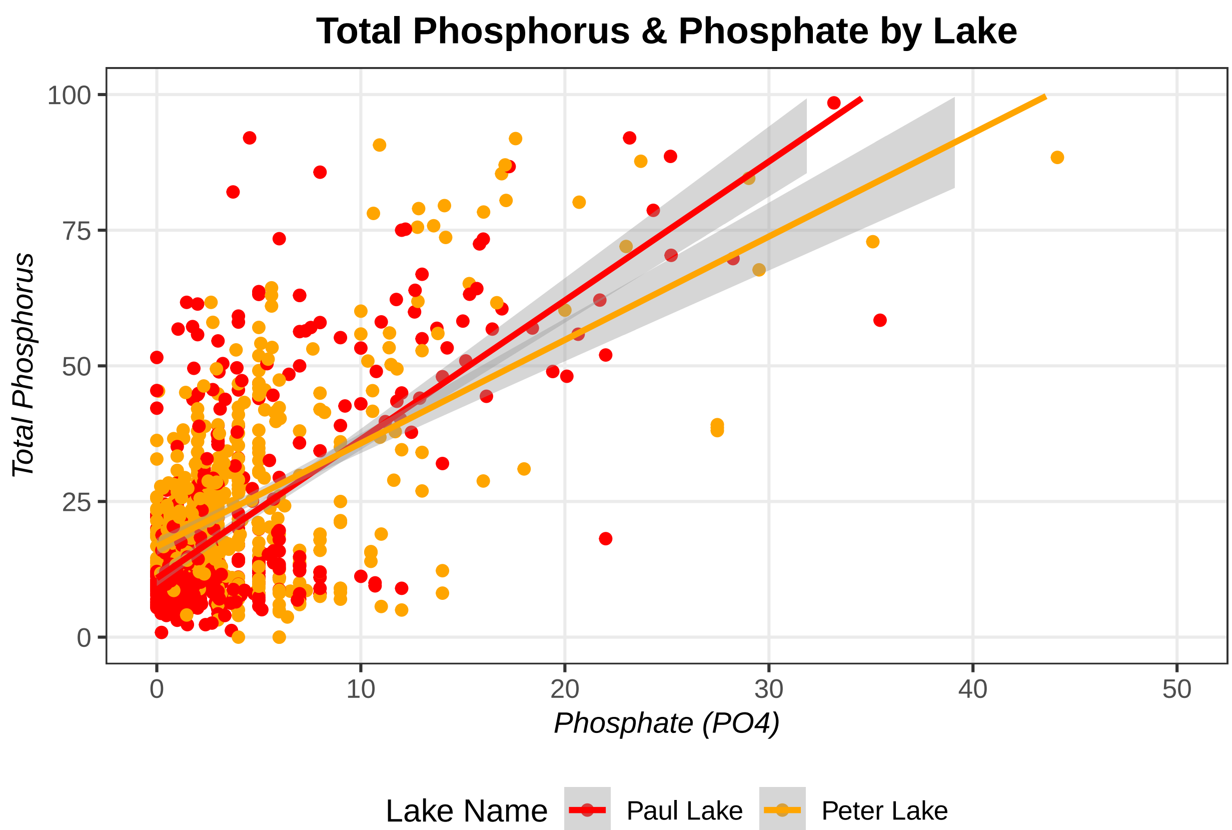
Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```r
#4
ggplot(NTL_LTER_PeterPaul, aes(x = po4, y = tp_ug, color = lakename)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  scale_color_manual(values = c("red", "orange")) +
  xlim(0, 50) +
  ylim(0, 100) +
  labs(
    title = "Total Phosphorus & Phosphate by Lake",
    x = "Phosphate (PO4)",
    y = "Total Phosphorus",
    color = "Lake Name"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 21964 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 21964 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_smooth()`).
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
NTL_LTER_PeterPaul <- NTL_LTER_PeterPaul %>%
  mutate(month = factor(month(sampledate, label = TRUE)))

# Plot A
temp_plot <- ggplot(NTL_LTER_PeterPaul, aes(x = month, y = temperature_C, fill = lakename)) +
  geom_boxplot() +
  labs(x = NULL, y = "Temperature (°C)", fill = "Lake") +
  theme_minimal() +
  theme(legend.position = "none")

# Plot B
tp_plot <- ggplot(NTL_LTER_PeterPaul, aes(x = month, y = tp_ug, fill = lakename)) +
  geom_boxplot() +
  labs(x = NULL, y = "Total Phosphorus", fill = "Lake") +
  theme_minimal() +
```

```
  theme(legend.position = "none")

# Plot C
tn_plot <- ggplot(NTL_LTER_PeterPaul, aes(x = month, y = tn_ug, fill = lakename)) +
  geom_boxplot() +
  labs(x = "Month", y = "Total Nitrogen", fill = "Lake") +
  theme_minimal() +
  theme(legend.position = "none")

# Extract the legends
legend_plot <- ggplot(NTL_LTER_PeterPaul, aes(x = month, y = temperature_C, fill = lakename)) +
  geom_boxplot() +
  labs(fill = "Lake") +
  theme_minimal() +
  theme(legend.position = "bottom")
legend <- get_legend(legend_plot)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

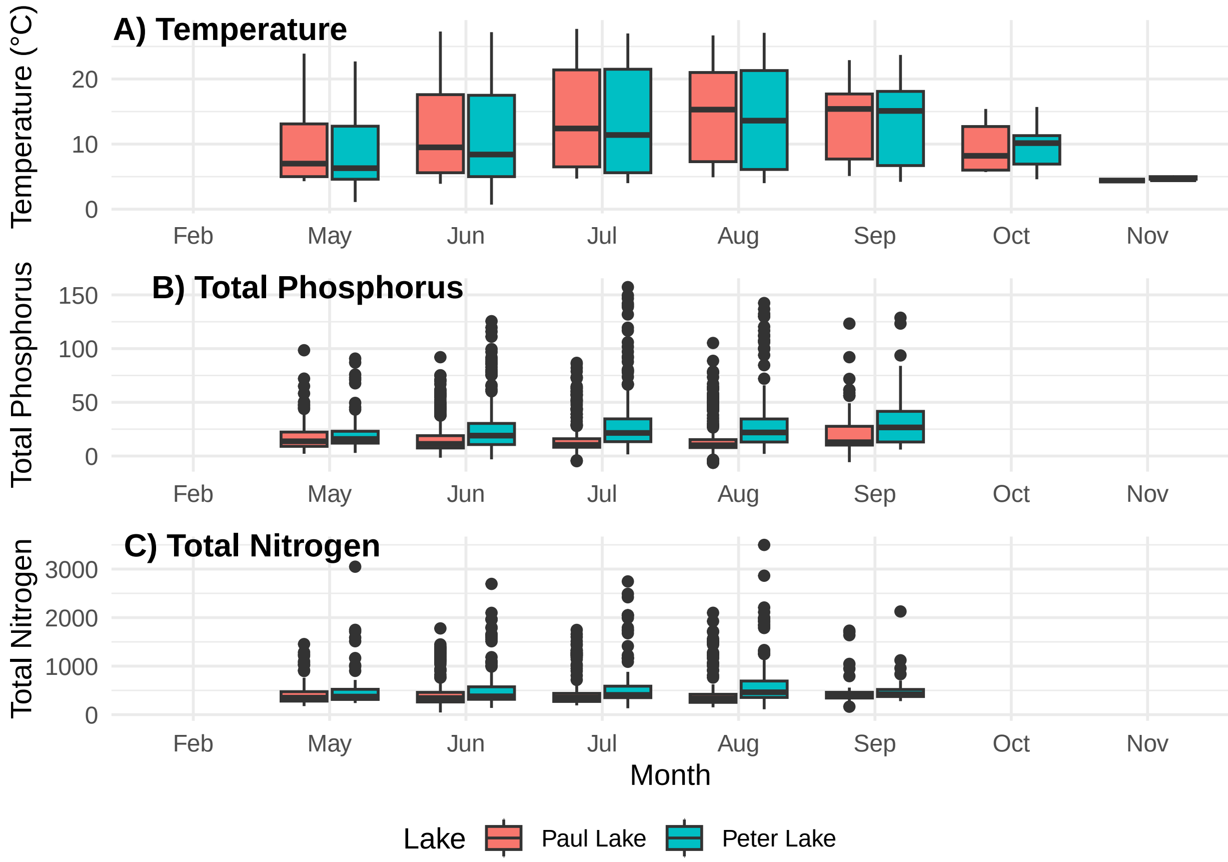
```
# Combine all the plots A, B, and C
combined_plot <- plot_grid(
  temp_plot, tp_plot, tn_plot,
  ncol = 1,
  align = "v",
  labels = c("A) Temperature", "B) Total Phosphorus", "C) Total Nitrogen"),
  label_size = 12,
  rel_heights = c(1, 1, 1.1)
)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
# Create legend
final_plot <- plot_grid(combined_plot, legend, ncol = 1, rel_heights = c(1, 0.08))
final_plot
```



Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer: The plots shows the lakes have th elowe temperature in the winter months and high temperature in the summer months. Although for total phosporous and nitrogen levels are different. I assume because one lake is a control lake.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)

7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

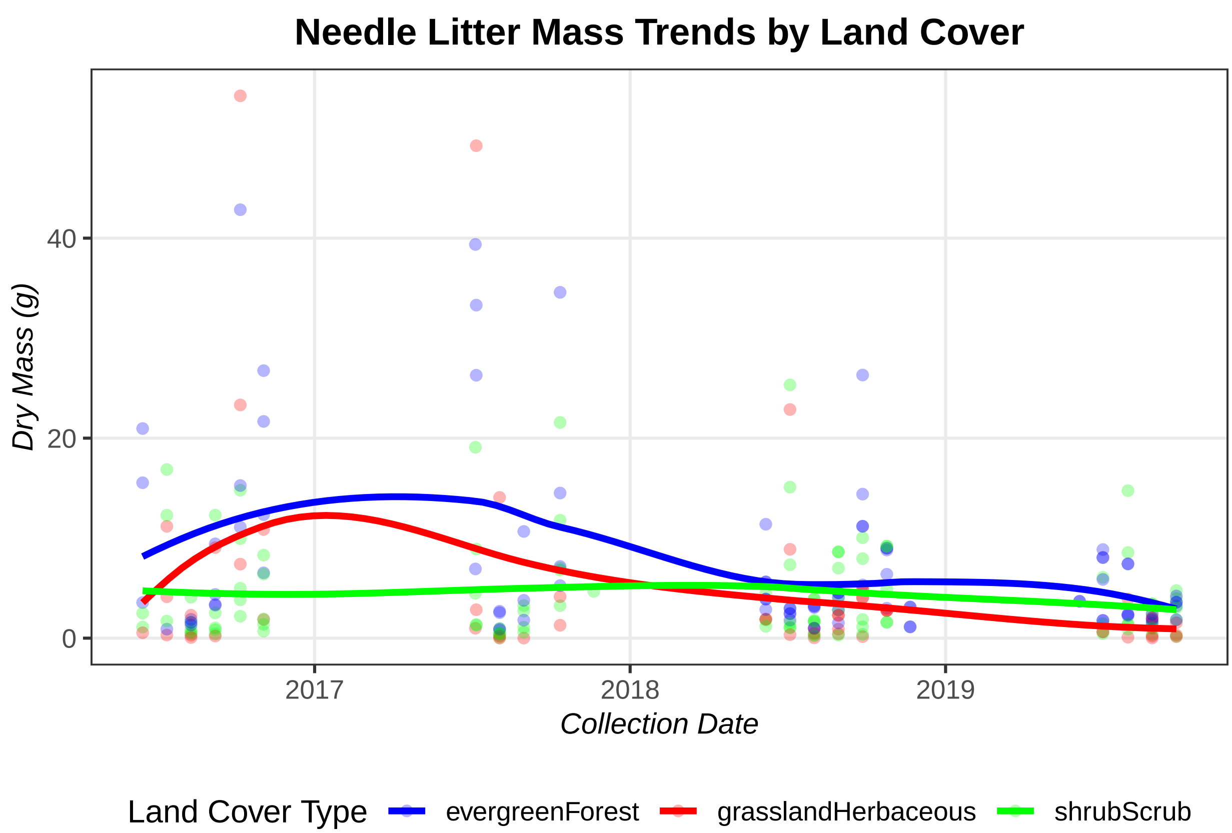
```r
needle_data <- Niwot_Ridge %>%
  filter(functionalGroup == "Needles")

# 6
plot_color <- ggplot(needle_data,
                     aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_point(alpha = 0.3, size = 1.5) +
  geom_smooth(method = "loess", se = FALSE, size = 1.2) +
  scale_color_manual(values = c("blue", "red", "green")) +
  labs(
    title = "Needle Litter Mass Trends by Land Cover",
    x = "Collection Date",
    y = "Dry Mass (g)",
```

```
    color = "Land Cover Type"
  )
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
print(plot_color)
```

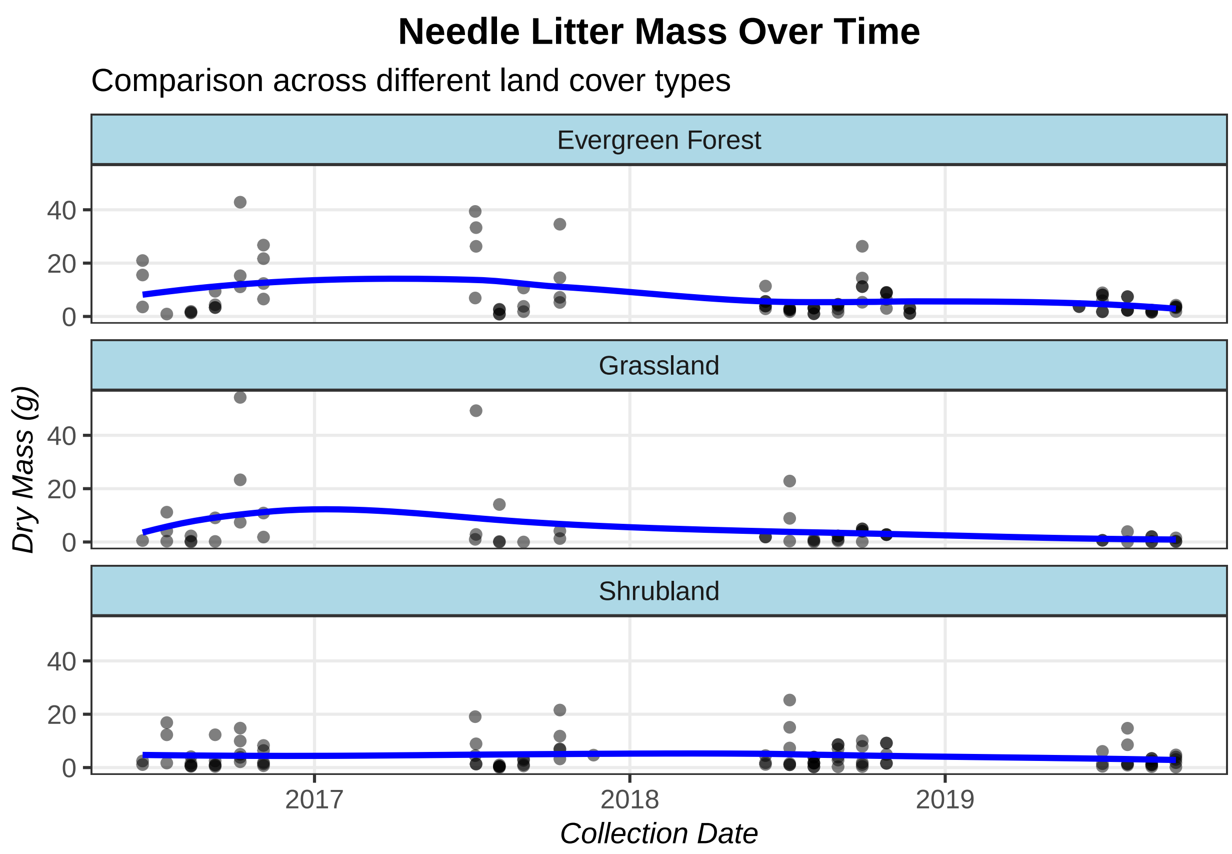
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# 7
plot_facet <- ggplot(needle_data,
                     aes(x = collectDate, y = dryMass)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  facet_wrap(~nlcdClass, ncol = 1,
             labeller = as_labeller(c(
               "evergreenForest" = "Evergreen Forest",
               "grasslandHerbaceous" = "Grassland",
               "shrubScrub" = "Shrubland"))) +
  labs(
```

```r
    title = "Needle Litter Mass Over Time",
    subtitle = "Comparison across different land cover types",
    x = "Collection Date",
    y = "Dry Mass (g)"
  ) +
  theme(strip.background = element_rect(fill = "lightblue"))
print(plot_facet)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Added theme above for aesthetic.
```

Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer: The plot 7 is more effective because it is more organized and show clear patterns with very low and less busy graphs.